

Carton Box Manufacturing System

Inventory Management, Job Card & Production Control – Requirement Document

1. Purpose

This document defines the requirements for a **Carton Box Manufacturing Management System** that manages **raw material inventory (paper rolls)**, **job card-based production**, **raw material consumption calculation**, **finished goods inventory**, and **controlled inventory movements with approvals**.

The system is intended for **small to mid-sized carton box manufacturers** seeking better production planning, inventory control, and operational transparency.

2. Business Objectives

- Maintain accurate raw material inventory (paper rolls)
 - Control material consumption during carton box production
 - Automate job card creation and material requirement calculation
 - Track finished carton boxes as inventory
 - Enforce approval workflows for inventory movement
 - Reduce wastage, stock mismatch, and production delays
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3. Scope

In Scope

- Raw material (paper roll) inventory management
- Job card-based production process
- Automated raw material requirement calculation
- Finished goods inventory management
- Inventory in/out controls with approvals
- Reports and stock visibility

Out of Scope (Phase 1)

- Machine-level integration
- Advanced costing and profitability

- GST invoicing
 - Accounting / ERP integration
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4. User Roles

4.1 Admin

- Master data management
- User & role configuration
- Approval workflow configuration

4.2 Production Manager

- Create and manage job cards
- Approve raw material consumption
- Monitor production status

4.3 Store / Inventory Manager

- Manage raw material stock
- Execute inventory in/out movements
- Request inventory approvals

4.4 Supervisor

- Validate production completion
 - Confirm finished goods quantity
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5. Functional Requirements

5.1 Raw Material Management (Paper Rolls)

Raw Material Master

- Create and manage paper roll types
- Attributes:
 - Material Code
 - Paper Type (Kraft / Duplex / etc.)
 - GSM
 - Roll Width
 - Roll Length / Weight
 - Supplier

Raw Material Inventory

- Maintain stock quantity per roll
 - Track:
 - Opening stock
 - Stock received
 - Stock issued
 - Stock balance
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5.2 Inventory Inward (Raw Material)

- Record raw material receipt
 - Capture:
 - Supplier
 - Purchase reference
 - Quantity received
 - Date of receipt
 - Inventory inward requires:
 - Store Manager entry
 - Production Manager / Admin approval
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5.3 Job Card Management (Production Process)

Job Card Creation

- Create job card for carton box manufacturing
- Job card attributes:
 - Job Card Number
 - Box Type / Size (L x W x H)
 - Ply Type
 - Quantity to Produce (X boxes)
 - Planned Start & End Date

Raw Material Requirement Calculation

System must automatically calculate: - Paper required per box (based on dimensions & ply) - Total raw material required for X boxes - Display required paper roll quantity

Example: - 1 box requires 2.5 sq.m paper - Order quantity = 1,000 boxes - Total paper required = 2,500 sq.m

System validates availability before job card approval.

5.4 Raw Material Issue Against Job Card

- Issue raw material against approved job card
- Stock deduction from raw material inventory
- Partial issue allowed

Controls

- Issue request raised by Store Manager
 - Approval required from Production Manager
 - System logs:
 - Issued quantity
 - Date
 - Issued by / Approved by
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5.5 Production Tracking

- Job card status tracking:
 - Created
 - Approved
 - In Production
 - Completed
 - Record actual production quantity
 - Capture wastage (optional field)
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5.6 Finished Goods Inventory (Carton Boxes)

Finished Goods Master

- Carton box type
- Size & specification

Finished Goods Stock

- Automatically add stock on job card completion
 - Track:
 - Quantity produced
 - Quantity rejected
 - Net finished goods added
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5.7 Finished Goods Inventory Movement

Stock Inward

- Auto-inward from production
- Supervisor confirmation required

Stock Outward

- Dispatch / sale / transfer
- Requires approval

Controls & Approvals

- Store Manager initiates
 - Production Manager / Admin approves
 - Audit trail mandatory
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5.8 Inventory Adjustments

- Manual stock adjustment (shortage / damage)
 - Mandatory reason capture
 - Admin approval required
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5.9 Reports & Dashboards

Inventory Reports

- Raw material stock report
- Finished goods stock report
- Low stock alerts

Production Reports

- Job card summary
 - Material consumption vs planned
 - Wastage analysis
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6. Approval & Control Mechanism

- Configurable approval workflows
- All inventory movements must:
 - Be role-restricted
 - Require approval

- Maintain audit logs
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7. Non-Functional Requirements

Performance

- Support up to 500 active job cards
- Inventory updates in real time

Security

- Role-based access control
- Action-level permissions

Auditability

- Full transaction history
 - Immutable audit logs
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8. Technology (Suggested)

Frontend

- React (Web application)

Backend

- Spring Boot or FastAPI

Database

- PostgreSQL

Deployment

- Dockerized deployment
 - On-prem or cloud
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9. Assumptions

- Material calculation formula is configurable
 - One unit of finished product = one carton box
 - Production wastage is recorded manually
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10. Future Enhancements

- Machine integration
 - Costing & profitability reports
 - Barcode / QR-based inventory
 - ERP integration
 - AI-based wastage prediction
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11. Acceptance Criteria (Sample)

- Job card cannot be approved without sufficient raw material
 - Raw material stock updates correctly after issue
 - Finished goods stock updates on production completion
 - Inventory movement requires approval
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